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Arrays of Networked Standard Geothermal Wells

1. LEVELIZED COST OF ELECTRICITY

We have chosen levelized cost of electricity (LCOE) as our Key Performance Indicator (KPI). LCOE is based on the time-value of money. It discounts expenditures and revenues to their "present value" based on how big they are and when they occur. To improve LCOE, you want to cut costs, but also push expenditures out and pull revenues in. What is our next LCOE plateau, at which new cooperating standards might converge?

Energy cannot just be clean. It must also be cheap. Here are some current energy cost estimates that suggest an LCOE goal of <1¢/kWh might be a good next cost plateau. From June 2025 new-build LCOE estimates from Lazard v18.0:

Technology	High Typical ¢/kWh from $/MWh
Solar PV (utility-scale)	7.8
Wind (onshore)	8.6
Geothermal (new)	10.9 – take this as the status quo
Nuclear (new)	22.0
Coal (new)	17.3
Gas combined cycle	10.9
Gas peaker (new)	25.1

Stretch Goal: Geothermal array LCOE under a penny per kilowatt hour (<1¢/kWh).

2. CAPACITY FACTOR

Capacity factor (CF) is a measure of how much a power plant actually produces compared to how much it might produce if run at full power ("nameplate") all the time.

CF = Energy over period ÷ Energy run at nameplate power for same period

Here are some rough CFs for various sources of energy, according to USDOE Office of Nuclear Energy:

Energy Source	Capacity Factor
Solar PV	23.4%
Geo	65.0%
Wind	34.3%
Nuclear	92.3%
Coal	42.6%
Gas combined cycle	59.9%

Gas simple	17.2%

Solar and wind CFs are low and the current quest is for “firm” power, using all sorts of energy storage devices. Geothermal has plenty of CF. In particular, geothermal energy is not affected by Earth’s rotation or cloudy days or when the wind doesn’t blow. Geothermal arrays network wells so that they can back up one another, driving CF toward 100%.

3. BASELINE WELL

We asked ChatGPT to design a geothermal well that is deep, coaxial, closed, and (for now) frack free. Here is the actual prompt:

“Dear ChatGPT, please help me design a geothermal well, one suitable for later building out a geothermal array. Drill one borehole with an injection pipe and production pipe (pipe-in-pipe). Be sure the well is closed with a turnaround at the bottom and exchanger at the top.

Use typical parameters for drilling and operating the well. Focus on the flow rate of the circulating working fluid. Record and report the various steady-state temperatures of note, for example top of production, bottom of injection. Do not use vacuum insulation (VIT) or conductive cement (CC) -- we'll try that next. After simulating the well, report the names and values of all parameters used. Most importantly, compute and report the levelized cost of electricity. Let the well's power output (MW) be supportable by a typical ORC. Let all reporting be metric. And let our goal be LCOE <1¢/kWh. Go!”

The well that she (GPT) came up with did not meet our LCOE goal:

3.1 Baseline Deep, Coaxial, Closed, Frack-Free Well Parameters

To see how much geothermal knowledge ChatGPT has accumulated, we simply asked her to design a well that is deep, coaxial, closed, and (for now) frack free.. She (GPT) came up with this verbatim:

Subsurface (Borehole)

- A closed, frack-free borehole contains a coaxial pipe-in-pipe system.
- The outer annular pipe carries cooler working fluid downward from the surface.
- The inner pipe carries heated working fluid upward toward the surface.
- Heat is absorbed conductively from surrounding rock along much of the borehole.

Bottom Turnaround

- At the bottom of the borehole, a U-turn assembly redirects the downward-flowing fluid from the outer pipe into the inner pipe.
- No mixing with formation fluids; the loop remains fully sealed.

Surface Completion

- The hot upflow enters a surface heat exchanger.
- Heat transferred to thermal-to-electric ORC, whose turbine drives a generator.
- ORC generator output passes through inverter, delivers DC power to well’s micro-grid

Waste Heat Rejection

- Heat not converted to electricity is rejected via air-cooled fans and fins mounted atop the surface completion

Return Flow

- After cooling in the heat exchanger, the working fluid is routed back into the outer downward pipe, closing the loop and restarting the cycle.

Parameter	Value
Depth	5 km
Gradient	25 °C/km

Mass flow	50 kg/s
Thermal power	14–16 MWth
Net power	1–1.5 MWe
Capacity factor	90–95%
LCOE	11.2 ¢/kWh

Next, we tried a quick improvement of the LCOE of our baseline well. We suggested that GPT use conductive cement (CC) to increase heat entering circulation toward the bottom of the well. We suggested that GPT use vacuum insulated tubing (VIT) to reduce heat loss toward the top of the well. And to assume that drilling costs are now half (50%) of what GPT had assumed, which is what drilling researchers are claiming.

LCOE was improved to 5.64 ¢/kWh.

Here are some details GPT disclosed:

Produced temperature at surface, Tprod,top(°C)

Depth (km) \ Flow (kg/s)	20	50	80	100
3	87.1	74.0	65.5	61.7
5	139.1	125.0	109.4	101.3
7	189.8	179.5	161.5	150.2
10	265.0	259.9	244.0	231.2

Net electric power, Pnet(MW_e)

Depth (km) \ Flow (kg/s)	20	50	80	100
3	0.45	0.73	0.71	0.61
5	0.96	1.93	2.27	2.27
7	1.46	3.22	4.16	4.41
10	2.19	5.12	7.16	7.99

LCOE (¢/kWh)

Depth (km) \ Flow (kg/s)	20	50	80	100
3	18.55	13.18	14.29	17.13
5	12.55	8.18	7.89	8.30
7	10.77	6.91	6.43	6.55
10	9.63	6.21	5.65	5.64

GPT gave the following advice after presenting the improved baseline well data:

To get to < 1¢/kWh, says GPT

Incremental well tuning alone will not get you to <1 ¢/kWh. The remaining gap must be closed by array-level effects:

GPT says you still need some combination of:

- More than 2× reduction in total capex (beyond halving drilling), shared infrastructure
- More than ~5× increase in net MW per well at similar capex, and/or
- Much cheaper ORC modules than $2500/kW, Mass-manufactured ORC module
- Array effects that cut per-well overheads dramatically (shared infrastructure, standardized install, learning curves)
- Learning curves on drilling
- Networked headroom & near-100% array CF
- Deferred drilling / staged deepening

Accounting for LCOEs associated with coal, gas, wind, solar, and geo is difficult. For example, some LCOEs include land use, others don't. Some LCOEs include fuel extraction, refining, and transport, others do not. Some LCOEs include capacity factors others do not. At the end of the day, says GPT, when you fold in land and fuel costs, capacity factors, and system-level considerations, geothermal – if you hit those drilling cost targets – can realistically push toward that <1¢/kWh goal. Fossil fuels still often come in cheaper in the short run, but geothermal arrays, with no fuel and nearly 100% capacity factor, could break the mold. Ultimately, reaching LCOE <1¢/kWh would put geothermal arrays in a category of their own.

Okay, let's explore. GPT says: To get LCOE <1¢/kWh you have to "turn every knob" and you must stop asking a single plain coaxial borehole to behave like a high-power heat exchanger. The winning combo is:

1. hotter rock (depth/gradient),
2. much higher effective heat-transfer (UA) (CC + geometry that increases area),
3. VIT near the top,
4. big array economics (shared T2E plant/cooling/switchgear), and
5. very low drilling $/m (your 50% claim, and then some learning-curve scaling).

Below is one internally consistent "All Improvements" baseline that lands at ~0.59 ¢/kWh. It's aggressive, but it shows exactly what has to be true.

4. "ALL IMPROVEMENTS" DEEP, COAXIAL, CLOSED-LOOP, FRACK-FREE, ARRAY-READY

Geometry & rock

- True vertical depth: 8.0 km
- Geothermal gradient: 25 °C/km
- Surface rock temp: 15 °C
- Rock temp at 8 km (estimate): 15 + 25·8 ≈ 215 °C
- Closed-loop coaxial: downflow in annulus, upflow in inner pipe
- Bottom turnaround: sealed U-turn
- No formation fluid production; no hydraulic fracking

Heat-transfer enhancements (still "closed")

- VIT length (top section): 2.0 km (minimize parasitic heat loss on the way up / down)
- Conductive cement (bottom section): 2.0 km (maximize formation→fluid conduction where rock is hottest)
- "Radiator section" at bottom (still sealed): an enlarged OD / high-conductivity liner / increased surface area section over the bottom ~1 km
  This is the non-negotiable part for big power in a closed loop: increase effective area/UA.

Working fluid & flow

- Working fluid: water (or water-like brine in a sealed loop)
- Mass flow: 90 kg/s (higher than your baseline 50 kg/s)
- cp (assumed): 4.2 kJ/kg-K

Steady-state temperatures (illustrative but consistent)

- Injection temp entering downflow: 50 °C

- Bottom downflow (after geothermal pickup): ~210 °C
- Production temp at surface (after VIT-limited losses): ~200 °C
- ΔT available across surface heat exchanger: ~150 K (200 → 50 °C)

Power

- Thermal power delivered to surface (estimate): $\dot{Q} \approx \dot{m} c_p \Delta T = 90 \cdot 4.2 \cdot 150 \approx 56.7$ MWth$\dot{Q} \approx \dot{m} c_p \Delta T = 90 \cdot 4.2 \cdot 150 \approx 56.7$ MWth
- Net T2E efficiency (air-cooled, large unit, good source temp): ~18% net
- Net electric power: 0.18 · 56.7 ≈ 10.2 MWe (call it 10 MWe net)

Operations/finance

- Capacity factor: 97% (closed-loop, dispatchable, lots of redundancy in arrays)
- Plant life: 30 years
- Real WACC: 6%

4.1 Cost model that matches "all improvements" (per well, but array-allocated)

This is where array thinking matters: you do not buy a tiny bespoke ORC and cooling stack per well. You allocate a share of a central, mass-produced T2E + cooling + electrical BOS.

Per-well CapEx allocation (aggressive, but what you asked for):

- Drilling + casing/completion (8 km, learning-curve, 50%+ reduction): $2.4M
- VIT incremental cost (top 2 km): $0.6M
- CC / conductive liner / radiator section: $0.9M
- Surface plant + cooling + electrical BOS (array-shared allocation): $1.5M
- Total CapEx per well: $5.4M

Fixed O&M: 2% of CapEx/year → $108k/yr

LCOE calculation (result: 0.59 ¢/kWh)

Annual energy:

$E = 10 \text{ MW} \cdot 8760 \cdot 0.97 = 84.972 \text{ GWh/yr}E = 10 \text{ MW} \cdot 8760 \cdot 0.97 = 84.972 \text{ GWh/yr}$

Capital recovery factor (6%, 30y):

- CRF ≈ 0.07265

Annualized capital:

- $5.4M \cdot 0.07265 = \$392k/yr5.4\text{M} \cdot 0.07265 = \$392\text{k/yr}$

Total annual cost:

- $392k + 108k = \$500k/yr392\text{k} + 108\text{k} = \$500\text{k/yr}$

LCOE:

- $\$500k/84.972GWh = \$0.00589/kWh = 0.589$ ¢/kWh$\$500\text{k}/84.972\text{GWh} = \$0.00589/\text{kWh} = 0.589$ ¢/kWh

Below 1 ¢/kWh (with these combined assumptions)

4.2 What this really implies (the non-sugarcoated part)

You don't get <1 ¢/kWh by "CC + VIT + cheaper drilling" alone if the well stays ~1 MWe. You get <1¢/kWh by making each well ~10 MWe net and keeping CapEx per well in the single-digit millions via:

- hotter rock (depth/gradient),
- a genuine bottom "radiator" section (big UA),
- centralized, shared T2E/cooling/BOS (array economics),
- learning-curve drilling.

If any one of those fails badly, you pop back above 1 ¢/kWh.

4.3 Quick stress test

Using the same energy output:

- If total CapEx/well rises $5.4M → $10M, LCOE becomes about 1.1 ¢/kWh (barely misses).
- If net power is only 5 MWe (everything else same), LCOE roughly doubles to ~1.2 ¢/kWh.

So the "knife edge" is: must be multi-MWe per well, not 1–1.5.

Engineered pathway: W1 → W2 → W3 to reach <1 ¢/kWh

It keeps your original framing (deep, coaxial, closed-loop; frack-free for now) and shows exactly what must change to get from 11.2 ¢/kWh to <1 ¢/kWh

4.4 Well concept (common to all cases)

We model a sealed, closed-loop coaxial well with a bottom turnaround (U-turn) and a surface heat exchanger coupled to a T2E (air-cooled ORC). Working fluid circulates continuously:

- Downflow: cooler fluid in the annulus (outer pipe / annular space)
- Upflow: heated fluid in the inner production pipe
- Bottom: sealed turnaround redirects flow from annulus to inner pipe
- Surface: hot upflow transfers heat to T2E; cooled fluid reinjected down annulus
- No formation fluid production and no mixing with formation fluids

4.5 Performance model (transparent and simple)

Thermal power delivered to surface is approximated by:

$\dot{Q}_{surf} \approx \dot{m}\, cp\,(T_{prod,surf} - T_{inj,surf})$ $\dot{Q}_{surf} \approx \dot{m}\, cp\,(T_{prod,surf} - T_{inj,surf})$

Net electric power:

$P_{net} \approx \eta_{net}\, \dot{Q}_{surf}$ $P_{net} \approx \eta_{net}\, \dot{Q}_{surf}$

Annual electric energy:

$E_{yr} = P_{net}\, 8760\, CF$ $E_{yr} = P_{net}\, 8760\, CF$

Levelized cost of electricity:

$LCOE \approx CRF \cdot CapEx + OpEx_{Eyr}$ $LCOE \approx CRF \cdot CapEx + OpEx_{Eyr}$

Where the capital recovery factor:

$CRF = \frac{r(1+r)^N}{(1+r)^N - 1}$ $CRF = \frac{r(1+r)^N}{(1+r)^N - 1}$

4.6 Case definitions and parameter tables (metric)

W1 — Baseline deep, coaxial, closed-loop, frack-free (no VIT, no CC)

W1 parameter table

Category	Parameter	Value	Notes

Geology	Depth	5,000 m	Vertical depth
Geology	Geothermal gradient	25 °C/km	Representative
Geology	Surface rock temperature	15 °C	Representative
Fluid	Working fluid	Water (sealed loop)	cp assumed constant
Fluid	Mass flow rate	50 kg/s	Your baseline
Completion	VIT length	0 m	none
Completion	Conductive cement length	0 m	none
T2E	Net conversion efficiency	~10%	low source temperature
Finance	Capacity factor (CF)	0.93	90–95% baseline
Finance	Lifetime	30 y	
Finance	Real discount rate (WACC)	6%	

W1 steady-state temperatures (illustrative)

Temperature point	Symbol	Value
Injection at surface (to annulus)	$Tinj,surf$ Tinj,surf	50 °C
Production at surface (from inner pipe)	$Tprod,surf$ Tprod,surf	~120 °C
Rock at bottom (5 km)	$Trock,5km$ Trock,5km	~140 °C

W1 power + economics (matches your stated baseline outcome)

- Thermal to surface: $\dot{Q}\approx 50\cdot 4.2\cdot(120-50)\approx 14.7\dot{Q} \approx 50\cdot 4.2\cdot(120-50)\approx 14.7$ MWth
- Net electric: $Pnet\approx 0.10\cdot 14.7\approx 1.5$Pnet $\approx 0.10\cdot 14.7\approx 1.5$ MWe
- Reported LCOE: 11.2 ¢/kWh (your baseline)

Interpretation: At ~140 °C rock, the closed-loop well is both temperature-limited (low T2E efficiency) and heat-transfer limited (limited UA), yielding ~1 MWe-class output and high LCOE.

W2 — Incremental improvements: add CC + VIT + 50% drilling cost

W2 keeps the same basic concept but adds:

- CC in the lower well section to increase effective formation→fluid conduction
- VIT in the upper well section to reduce parasitic losses to cooler formations
- 50% drilling cost reduction per your "researchers claim" assumption

W2 parameter changes (relative to W1)

Parameter	W1	W2
VIT length	0 m	~1,500–2,000 m
Conductive cement length	0 m	~1,000–2,000 m
Drilling cost	1.0×	0.5×
Net electric	~1–1.5 MWe	higher (modestly)
LCOE	11.2 ¢/kWh	5.64 ¢/kWh (your result)

Interpretation: W2 demonstrates meaningful gains, but remains a single-digit MWe challenge unless we increase bottom temperature and/or effective UA much more.

W3 — "All improvements" engineered to cross <1 ¢/kWh

W3 is the first case that is explicitly engineered to make the LCOE target feasible. It combines:

1. Deeper/hotter rock (to lift T2E efficiency)
2. VIT (to preserve produced temperature)
3. CC (to lift conduction where rock is hottest)
4. Bottom "radiator section" (still sealed; not fracking) to increase effective heat-transfer area/UA
5. Array economics (shared T2E/cooling/BOS rather than bespoke per-well plant)
6. Aggressive drilling learning curve (beyond the initial 50%)

W3 parameter table

Category	Parameter	Value	Notes
Geology	Depth	8,000 m	deep/hotter
Geology	Gradient	25 °C/km	as before
Geology	Surface rock temperature	15 °C	
Fluid	Working fluid	Water (sealed loop)	cp ≈ 4.2 kJ/kg-K

Fluid	Mass flow rate	90 kg/s	increased for power
Completion	VIT length (top)	2,000 m	reduce losses
Completion	CC length (bottom)	2,000 m	increase conduction
Completion	Bottom radiator section length	1,000 m	enlarged OD / high-k liner / increased area
T2E	Net conversion efficiency	18% net	larger plant; higher source temp
Finance	Capacity factor (CF)	0.97	array + redundancy
Finance	Lifetime	30 y	
Finance	Real WACC	6%	

W3 steady-state temperatures (engineered targets)

Temperature point	Symbol	Value	Notes
Injection at surface	$Tinj,surf$Tinj,surf	50 °C	after T2E HX
Bottom rock temp (8 km)	$Trock,8km$Trock,8km	~215 °C	15 + 25·8
Production at surface	$Tprod,surf$Tprod,surf	~200 °C	VIT preserves heat
Effective ΔT across surface HX	ΔTΔT	150 K	200 → 50

W3 thermal power and net power

Thermal power to surface:

$Q˙surf$≈90·4.2·150≈56.7 MWthQ˙ surf ≈90·4.2·150≈56.7 MWth

Net electric:

$Pnet$≈0.18·56.7≈10.2 MWePnet ≈0.18·56.7≈10.2 MWe

Annual energy:

Eyr=10.0·8760·0.97≈84.97 GWh/yEyr =10.0·8760·0.97≈84.97 GWh/y

W3 cost allocation (array-shared)

To avoid the "one well = one bespoke ORC" trap, W3 allocates a share of centralized, standardized surface plant.

Cost element (per well allocation)	Value (USD)	Notes
Drilling + casing/completion	$2.4M	deep drilling learning curve
VIT increment (top 2 km)	$0.6M	
CC + radiator section	$0.9M	CC + enhanced heat-transfer geometry
Surface plant + cooling + BOS allocation	$1.5M	shared across array
Total CapEx per well	$5.4M	

O&M (fixed):

- 2% of CapEx/year ⇒ $108k/year

Finance:

- Real WACC rr=6%, N=30 → CRF ≈ 0.07265

Annualized capital:

- 5.4M·0.07265≈392k/yr$5.4M·0.07265≈$392k/yr

Total annual cost:

- 392k+108k=500k/yr$392k+108k=$500k/yr

W3 LCOE

$LCOE$=500k84.97 GWh≈0.00589 $/$kWh$=0.589 ¢/$kWh$LCOE=84.97 GWh500k ≈0.00589 $/kWh=0.589 ¢/kWh

W3 achieves <1 ¢/kWh under the combined "all improvements" assumptions.

What must be true (design requirements, not wish-casting)

W3 only works if these engineered requirements are met:

1. Produced temperature near ~200 °C (needs deeper/hotter rock and minimized upper losses via VIT)
2. Closed-loop UA is increased substantially (CC helps, but the bottom radiator section is what enables multi-MWth pickup without fracking)

3. Net power per well is multi-MWe (≈10 MWe here)
4. Surface plant is shared and standardized (array economics)
5. Drilling CapEx per meter falls sharply with learning curves (and/or robotic drilling, deferred drilling, etc.)

Sensitivity

Because LCOE scales approximately with $CapExPnet \cdot CF$Pnet ·CFCapEx , W3 sits on a clear feasibility edge: if $Pnet$Pnet falls from 10 MWe to 5 MWe, LCOE roughly doubles to ~1.2 ¢/kWh; if CapEx rises from $5.4M to $10M at 10 MWe, LCOE rises to ~1.1 ¢/kWh. Thus, multi-MWe per well and array-shared surface economics are not optional—they are the principal enablers of <1¢/kWh.

W3 DESIGN SPEC

Deep, Coaxial, Closed-Loop Geothermal Well (Array-Optimized)
Target LCOE: ≤ 1.0 ¢/kWh (real, unsubsidized)

Purpose and scope

This specification defines a single standardized geothermal well module intended for deployment in large geothermal arrays. The well is:

- Deep
- Coaxial (pipe-in-pipe)
- Closed-loop (no formation fluid production)
- Frack-free (no hydraulic stimulation)
- Array-optimized (shared surface plant, shared BOS)

The design objective is to achieve multi-MWe net output per well with LCOE < 1 ¢/kWh, assuming mature drilling learning curves and array-scale deployment.

System architecture (fixed)

- Well type: Vertical, sealed, closed-loop
- Flow configuration:
  - Downflow in annulus
  - Upflow in inner production pipe
- Bottom: Sealed U-turn turnaround
- Surface: Heat exchanger feeding centralized T2E (air-cooled ORC)
- Cooling: Air-cooled (no consumptive water use)
- Electrical: DC output to array micro-grid

No formation fluid exchange occurs at any point.

Geological and thermal design parameters (fixed)

Parameter	Value	Units	Notes
True vertical depth	8,000	m	No laterals
Geothermal gradient	25	°C/km	Conservative
Surface rock temperature	15	°C	
Rock temperature at TD	215	°C	15 + 25×8

Well geometry and completion

Borehole

Parameter	Value
Nominal borehole diameter	311 mm (12.25 in)
Total drilled length	8,000 m

Coaxial tubing

Element	OD / ID	Length
Inner production pipe	Sized for ≤ 3 m/s upflow	Full depth
Outer annulus	Remaining bore ID	Full depth

(Material selection assumes high-temperature steel alloys suitable for 215 °C continuous operation.)

Thermal enhancements (mandatory)

Vacuum-insulated tubing (VIT)

- Location: Upper section
- Length: 2,000 m
- Purpose: Minimize parasitic heat loss to cooler formations

Conductive cement (CC)

- Location: Lower section
- Length: 2,000 m
- Thermal conductivity: ≥ 5 W/m-K (effective)
- Purpose: Increase formation-to-fluid heat flux

Bottom radiator section (sealed)

- Length: 1,000 m
- Features (one or more):
  - Enlarged OD liner
  - High-conductivity materials
  - Increased effective heat-transfer surface area
- Purpose: Increase effective UA without fracking or fluid exchange

This section is the primary thermal power enabler.

Working fluid and circulation

Parameter	Value	Units
Working fluid	Water (sealed loop)	—

Mass flow rate	90	kg/s
Specific heat (assumed)	4.2	kJ/kg-K
Circulation mode	Continuous	

Steady-state temperature targets (design requirements)

Location	Symbol	Value
Injection temperature (surface)	$Tinj,surf$	50 °C
Bottom fluid temperature	$Tbottom$	≈ 210 °C
Production temperature (surface)	$Tprod,surf$	≈ 200 °C
Effective ΔT at surface HX	ΔT	150 K

Upper losses after VIT are limited to ≤ 10 °C.

Thermal and electric performance (required)

Thermal power to surface

$Q˙surf=m˙cp\Delta T$=90·4.2·150≈56.7 $MWth$Q˙ surf =m˙cp ΔT=90·4.2·150≈56.7 MWth

T2E conversion

Parameter	Value
T2E type	Air-cooled ORC
Net conversion efficiency	18 %

Net electric output

$Pnet$=0.18·56.7≈10 MWePnet =0.18·56.7≈10 MWe

Operations and availability

Parameter	Value
Capacity factor	97 %
Dispatchability	Full

Operating life	30 years

Closed-loop operation enables high CF with minimal decline.

Cost specification (array-allocated, real USD)

Capital expenditure (per well allocation)

Component	Cost (USD)
Drilling + casing + completion	$2.4 M
VIT (2 km)	$0.6 M
CC + radiator section	$0.9 M
Surface plant + cooling + BOS (shared)	$1.5 M
Total CapEx per well	$5.4 M

Operating expenditure

- Fixed O&M: 2 % of CapEx/year = $108 k/year

Financial assumptions

Parameter	Value
Real discount rate (WACC)	6 %
Project life	30 years
Capital recovery factor	0.07265

LCOE calculation (requirement check)

Annual energy:

E=10·8760·0.97=84.97 GWh/yE=10·8760·0.97=84.97 GWh/y

Annualized capital:

5.4M·0.07265=392k \$/y5.4M·0.07265=392k \$/y

Total annual cost:

392k+108k=500k \$/y392k+108k=500k \$/y$LCOE$=500k84.97GWh=0.589 ¢/kWhLCOE=84.97GWh500k =0.589 ¢/kWh

Meets requirement: LCOE < 1 ¢/kWh

Design validity conditions (explicit)

This design fails the LCOE requirement if any of the following occur:

- Net electric output < 5 MWe
- Total CapEx per well > $10 M
- Produced surface temperature < 180 °C
- Capacity factor < 90 %
- No effective bottom radiator / UA enhancement

Status

W3 is a valid engineered baseline, not a proof of existence. It defines what must be built for sub-1 ¢/kWh geothermal using closed-loop arrays.

5. DEFERRED DRILLING, LEARNING CURVE ARRAYS

Array architectures unlock economic levers unavailable at the well level. Shared infrastructure reduces per-well overhead, while standardization enables learning rates of 10–20% per cumulative doubling in drilling and surface equipment. Deferred drilling reduces present-value cost. Together, these effects reduce LCOE into the 1–3 ¢/kWh range.

Illustrative LCOE Progression

Stage	Key Lever	Capacity Factor	LCOE (¢/kWh)
Standalone	Baseline	90–95%	5–10
Optimized	VIT+CC	90–95%	4–7
Array	Shared infra	95–100%	1–3

Drilling and completion dominate geothermal capital cost, commonly accounting for 60–75% of total installed cost. Surface power conversion equipment, electrical infrastructure, and site works comprise the remainder. This cost structure makes geothermal particularly sensitive to learning curve effects in drilling operations.

Historical data from oil and gas drilling and from geothermal pilot programs indicate learning rates in the range of 10–20% per cumulative doubling of drilled meters. When applied across arrays comprising hundreds of identical wells, these learning effects materially reduce average well cost over time.

Array deployment further reduces effective cost through shared infrastructure and through deferred drilling strategies. By postponing the deepest and most expensive segments of wells until performance is validated or demand increases, developers reduce discounted capital cost.

Representative Capital Cost Breakdown

Cost Element	Share of Total
Drilling and completion	60–75%
Power conversion equipment	10–15%
Electrical infrastructure	5–10%
Site works and roads	5–10%
Engineering and permitting	5–10%

6. SCALING AND DEPLOYMENT SCENARIOS

At modest scale, arrays comprising 20–50 wells produce tens of megawatts of firm power suitable for industrial loads or regional grids. At a larger scale, arrays of 100–200 wells approach gigawatt-class installations when deployed as arrays, while retaining modularity and fault tolerance.

Land requirements scale favorably. With well spacing of 150 meters, a 10×10 array occupies approximately 2–3 square kilometers, including access lanes and surface equipment. At 200 meters, ~1,000 acres might generate megawatts to a gigawatt. Array power density exceeds that of most renewable generation technologies while maintaining firm output.

These characteristics make geothermal arrays suitable for replication across diverse geographies, including regions without high-temperature hydrothermal resources.

7. JUMP-START PROTOCOL

A battery robot (2–5 MWh capacity, 2–4 MW power) energizes the DC micro-grid bus to spin up the pumps in 5–10 initial wells. When these wells reach full power, they can recharge the robot and start waiting idle pumps and their wells. The array reaches 80+ active wells within 60–120 minutes depending on ambient conditions and sequencing. With CF >95%, we anticipate only needing one "jump start" in the 30-year life of an array, that being the initial jump start. If and when an array well fails, the array will keep operating at full nameplate power while a repair robot navigates lanes to the failed well and gets it back on line. The standards for well subsystems must have field replaceable units (FRUs) to enable robot replacement of failed FRUs.

8. GEOTHERMAL STORAGE

One key development of the Internet was storage. Storage absorbs supply uncertainty, demand uncertainty, and switching uncertainty. The Internet started with almost zero storage, but as it grew, storage cropped up everywhere, in the switches, routers, servers, desktop PCs, and intelligent phones.

Geothermal storage is today a hot topic, just like on the Internet. The uses of energy are uncertain as are its sources. We will have to come back to this topic in later papers, but our starting point is that array micro-grids and a little well buffering are all we need to enable geothermal wells to back up one another and to provide power to bridge outages elsewhere. The robotic vans will have large batteries for further energy storage. And then there's always Earth, a very big heat energy storage that can use arrays to deliver high CF power. tion

A 10×10 array networks 100 standard wells arranged in rows and columns. So, how close together can the wells be spaced? How much land will arrays take?

Thermal interference analysis by GPT for our example well indicates optimal spacing of 150–300 m. Closer and the wells start stealing heat from one another. Further and you start wasting land and paying for longer lanes, trenches, pipes, and cables. The optimum array LCOE is flat across the range. GPT recommends starting with 200 m as a baseline.

The 10x10 array land is flat with 10 rows and 10 columns. Total land area: approximately 1.8 km × 1.8 km = 3.2 km2 = 800 acres. The TV show LANDMAN shows lots of flat land in Texas.

Separation of 200 m leaves plenty of room to avoid thermal interference among wells, to bury the micro-grid DC connections, to mount the ORC converters atop the well pipes, to vent waste heat safely upward through the fans and fins, and to leave lanes for robot vans to access wells and store materials. By choosing 200 meters separation, we make the array 800 acres.

9. DRILLING COST REDUCTION

Recent advances in drilling technology achieve a 50% reduction in drilling and completion costs. With drilling representing 70% of baseline CAPEX, total CAPEX reduces from $30M to approximately $19.5M.

Economic impact:

At 5 MW net: LCOE 5.27 ¢/kWh (85% CF)

At 5 MW with array CF 95%: LCOE 4.75 ¢/kWh

At 10 MW net: LCOE 2.63 ¢/kWh (85% CF)

10. DEFERRED DRILLING STRATEGY

We notice that LCOE computes present values, making it impactful to move expenses into the future and power production into the present. Reducing or just deferring drilling expenses can greatly reduce LCOE. And notice that since drilling expenses increase super linearly with well depth, the most future drilling expenses are also the most expensive. Hence, let's figure out how to defer deeper drilling needed decades in the future.

Deferred drilling operates by initially drilling to the minimum depth achieving ORC threshold inlet temperature, then operating until performance degrades below threshold. Wells are selectively deepened ((re)drilled) only when that reduces LCOE, leveraging:

1. Time-value discounting of future drilling expenditures
2. Learning curve cost reductions in drilling technology
3. Selective deepening based on individual well performance

Economic Impact: Net present value of drilling costs reduced by approximately 10–30% versus "all-at-once" strategies. Combined with drilling progress (−50% on drilling costs) and deferred strategy (−20% on remainder), effective CAPEX reaches $14–16M per well pad at volume production.

Illustrative LCOE with deferred drilling:

- 5 MW net, CF 95%: LCOE 5.8 ¢/kWh
- 10 MW net, CF 95%: LCOE 3.0 ¢/kWh

Micro-Grid Configuration: Underground DC power cables run along columns with cross-ties at rows, enabling:

- Automatic fault detection and isolation
- Instantaneous rerouting of power from reserve wells or array well headroom
- Networked control and sensing for optimization
- Array capacity factor toward 100% using N+ reserve strategy

The micro-grid network can connect wells that are rated above nameplate. This is called “headroom”. When wells fail, the micro-grids switch in headroom power from other wells operating above the nameplate. Investments in headroom increase CF. How much CF do you want?

Jump-Start Protocol: A battery robot (2–5 MWh capacity, 2–4 MW power) energizes the DC bus to spin up the pumps in 5–10 initial wells. When these wells reach full power, they can recharge the robot and start any idle well pumps and their wells. The array reaches 80+ active wells within 60–120 minutes depending on ambient conditions and sequencing. No diesel generation required. With CF >95%, we anticipate only needing one “jump start” in the 30-year life of an array, that being the initial jump start. If and when an array well fails, the array will keep operating at full nameplate power while a repair robot navigates lanes to the failed well and gets it back on line. The standards for well subsystems must have field replaceable units (FRUs) to enable robot replacement of failed FRUs.

Jump starting can be used during drilling. Not all the wells are dropped in at once. Early wells can power the drilling of later wells. Later wells can be deferred, their cost reduced by present value in LCOE.

Standardized Robot Economy: The array architecture separates well ownership from service provision. Standardized robots for trenching, drilling, make-up water delivery, and maintenance compete in an open market, driving costs down also via Wright’s Law learning curves.

11. VALUE OF FIRM POWER

So, what is capacity factor worth? Should we deduct the LCOE of a 95% CF from the array’s total LCOE?

Geothermal arrays with 95%+ capacity factors avoid the storage, overbuild, and transmission requirements of intermittent renewables. Conservative estimates value this "firmness" at 1.0–1.5 ¢/kWh in system planning.

Firm LCOE = Technical LCOE − Firmness Credit

Current credible near-term stack (6–8 MW net per well pad, CAPEX $16–20M, CF 95%):

- Technical LCOE: 2.8–4.5 ¢/kWh
- Firm LCOE: 1.3–3.5 ¢/kWh

12. PATHWAY TO <1¢/KWH

Achieving LCOE <1¢/kWh requires convergence of factors:

1. Array-scale capacity factor (95%): Amortizes fixed costs over maximum energy production
2. Higher net power per pad (10–20 MW): Through deeper wells, higher flow rates, or multiple parallel strings

3. Learning-curve CAPEX reduction: Standardized components, mass manufacturing, rapid drilling progress, and fierce competition driving per-well costs to $7–15M
4. Optimized thermal management: VIT/CC enhancements with improved ORC efficiency (12–14% net)
5. O&M efficiency: Maintaining costs 2% of CAPEX through standardization and robotics

Target Scenarios:

- 10 MW net, CF 95%, CAPEX $10M: LCOE 1.21 ¢/kWh
- 10 MW net, CF 95%, CAPEX $7M: LCOE 0.85 ¢/kWh
- 20 MW net, CF 95%, CAPEX $15–20M: LCOE 0.9–1.2 ¢/kWh

The analysis shows that LCOE <1¢/kWh geothermal electricity is technically feasible through systematic optimization rather than breakthrough technologies. Each enhancement – VIT, CC, flow optimization, deferred drilling, array networking, learning curves, innovation curves – contributes incremental improvements that compound into transformative economics.

ChatGPT summarizes these technical challenges:

- Achieving 10–20 MW net per well pad through deeper drilling, higher flow rates.
- Long-term reliability of VIT under thermal cycling
- Optimization of CC formulations for enhanced heat transfer
- Development of rapid, low-cost drilling technologies

13. EMISSION AND CAPACITY FACTOR CREDITS

A geothermal array does not emit pollutants (e.g., zero CO2) and has a high capacity factor CF, >95%. These major benefits are not included so far in array LCOE. So, what is the value of CO2 not emitted at 1MW? What is the value of a CF of 100%?

14. TO FRACK OR NOT TO FRACK: VALUE OF CLOSED FRACK

What if we use fracking after all? Closed fracking? What would be the increase in heat flow? What would be the reduction (if any) of array LCOE? There would still be no exchange of working fluids, just conduction plus increased convection in native fluids near the boreholes and fracks?

Closed fracking still looks like fracking to regulators and investors; it still carries and must explain away seismicity and well pollution risks.

15. LEARNING CURVE, WRIGHT'S LAW, INNOVATION CURVE

The pathway to LCOE <1¢/kWh relies heavily on learning curves in drilling and surface equipment manufacturing. Historical precedent from solar PV (10× cost reduction over 15 years) and wind (3× cost reduction over 20 years) suggests such trajectories are achievable with sufficient deployment volume.

The deferred drilling strategy leverages time-value of money and technological learning. By deferring expensive drilling into the discounted future, projects improve NPV while benefiting from continued cost reductions and potential technological improvements.

The proposed networked geothermal arrays offer competitive economics today (LCOE 1.3–3.5 ¢/kWh firm) with clear pathways to industry-leading costs LCOE <1¢/kWh) through continued optimization and scale.

Wright's Law says the cost of a manufactured product goes down a learning curve at a fixed rate with cumulative shipments. Looking at ORC efficiency, the array produces far more (waste) heat energy than the array produces electrical energy. And still, there is room atop the arrays for wind turbines. What shall we call an array producing geothermal electricity, recovered geothermal waste heat (MWth), photovoltaic DC, and wind power? Geothermal cogeneration. each doubling of production. This is one of the advantages of the Internet mindset. Many fewer mainframe geothermal wells are produced compared to the many standard small wells. The mainframes in various infrastructures produce at lower costs to start, but over time the higher numbers of smaller products, learning faster, overtake the mainframes. Surprises often arise when cheaper products enabled by Wright's Law increase demand, which further drives learning and Wright's Law again, and so on – positive feedback rides again. In the meantime, for those evangelizing the Internet mindset and Wright's Law, there's a lot of convincing to do.

What would be the LCOE if we took Wright's Law and applied it to well production with a learning rate of 20%? What would the new LCOE be?

Assume that our current baseline well's LCOE is ~3 ¢/kWh.

Apply Wright's Law to well production (learning rate (LR) = 20%)

Wright's Law (learning curve) cost multiplier after scaling cumulative production from N0to N:

Multiplier=LRlog2(N/N0)

If we treat your 100-well array as learning from 1 → 100 wells:

Doublings: log2(100)=6.644

For LR = 20%, multiplier =0.86.644=0.227

New LCOE (if LCOE scales ~proportionally with well cost): 3.0×0.227=0.68 ¢/kWh

Learning curve reductions of LCOE may be reaching. So too might be innovation curve reductions. The former are driven by shipments compounding, the latter by research compounding over time. Standards for geothermal arrays have to anticipate learning and innovation advances. And if done right, the standards will encourage all sorts of fierce competition, driving costs even further down.

And we are going to need more than a million wells in our 10,000 geothermal arrays.

16. GEOTHERMAL COGENERATION

It's hard not to notice that geothermal wells have large heat fans and fins and that they release considerable low-grade (low-enthalpy) heat up into the atmosphere. If the wells are only able to capture 10% of the well's heat in the making of electricity, where does the other 90% go? While such heat cannot be used to generate electricity, there is a lot of it and many uses for it directly, without T2E conversion. In fact, direct heat carries more energy than the electricity produced by the wells. Cogeneration of electricity and heat will be a future topic.

Artificial intelligence is not supposed to be creative, but our GPT noticed all by herself that arrays might re-use their land. She suggested we install photovoltaic panels over our geothermal wells. But then we noticed that a geothermal well produces a huge quantity of so-called waste heat. So, instead of wasting the heat by sending it upward into the sky through fans and fins on the roof of the wells, let's aggregate it horizontally above the lanes and send it to a nearby use of heat – district heating, industrial, agricultural.

17. REDRILLING

Array control is distributed among, say, the four corners of the micro-grid. Each corner has a substation interconnect and can back up the other three controls – redundant and coordinated. The controls cooperate to survey the wells and order pumps, valves, switches, and robot vans to keep the array constantly optimal. When failures occur, or are going to occur, the controls detect them and take action. This is how to offer a CF of nearly 100%.

Among the actions the controls can order are to increase a circulation pump to cover for a failed well or to order a robot to drill a new well to meet imminent increase of demand. Or the controls might order a new well based on a prediction it has made about looming well drawdowns. That new well might be in virgin territory, or say between two more productive wells.

(Re)drilling is an important technology. And it's among the practices of the oil and gas industries. (Re)drilling an abandoned well from long ago is common practice, but it is different from drilling a new well anticipating that it will be redrilled repeatedly in the future. And it is different from redrilling a well that was drilled to be redrilled. The well becomes a reusable platform.

If you want to re-drill later, it's best you decide up front, says GPT:

1. Casing program
    a. Leave a clean kickoff window
    b. Oversize upper casing
2. Cement choice
    a. Flexible / thermally tolerant cement
3. Verticality
    a. Straighter holes = easier re-entry
4. Instrumentation
    a. Fiber, temperature logs that survive re-entry
5. Array planning
    a. Wells spaced so deeper sidetracks don't collide

18. NEXT STEPS: SHOTS ON GOAL

We have entered a conversation with OpenAI's Chat GPT about the scaling of geothermal energy. GPT conclusions are to be taken with a grain of salt, and we are. We call our list of next steps what a ice hockey player would call shots on goal:

We have presented a framework for networked geothermal well arrays targeting LCOE <1¢/kWh electricity generation. GPT offers its findings:

1. Bare-bones 5 km wells yield 1–2 MW net. Achieving 10 MW per pad requires deeper/hotter.
2. VIT and CC enhancements provide 20–35% power uplift with modest CAPEX additions.
3. Drilling cost reductions of 50% enable LCOE of 2.6–5.3 ¢/kWh.
4. Optimal flow rates of 70–100 kg/s balance thermal throughput against temperature approach and pumping costs.
5. Deferred drilling strategies reduce effective CAPEX by 10–30% through NPV optimization.
6. 10×10 arrays with 200 m spacing and 95%+ capacity factors deliver firm LCOE of 1.3–3.5 ¢/kWh.
7. Pathways to <1¢/kWh exist through continued learning curves, standardization, and economies of scale

The proposed architecture -- geothermal arrays: standardized wells, modular components, networked micro-grids, and competitive robot economies -- enables rapid deployment and continuous improvement. With geothermal offering zero emissions, baseload reliability, and location flexibility, and low cost, these economics position it as a cornerstone technology for global decarbonization.

Future work will validate these projections through more rigorous Julia-based modeling, pilot demonstrations, and field deployments. The goal remains clear: abundant, firm, zero-emission (clean) electricity at costs enabling universal energy access (cheap).

Scenario	Net MW	CF	CAPEX	LCOE (¢/kWh)
Baseline 5 MW	5 MW	85%	$30M	8.1
VIT + CC	6.5 MW	85%	$32M	~7.1
50% Cheaper Drilling	10 MW	85%	$19.5M	2.6
Deferred Drilling + Learning	10 MW	95%	$14–16M	~3.0
Optimized Flow 100 kg/s	10 MW	95%	$7–10M	0.85–1.2

Well separation < 200 meters?

(Re)Drilling: Drilling later is cheaper than drilling now.

- Capital deferral
- Learning curve capture with shipments
- Innovation curve capture with research
- Thermal headroom
- Long-lived assets that *improve* with age

Cogeneration with TENs, standard port before fans and fins

Closed Fracking, Permeability

sCO_2 Working Fluid

Arrays: Capacity Factor, Jump Start, Deferred Drilling

Array Robots

Standards: trucks, carts, attachments

Control software at each corner, cooperating, backup

Detailed transient thermal modeling in Julia (See Wittenstein et al)

Site-specific geological and thermal property measurements

Pilot demonstrations of VIT and CC technologies
Field trials of deferred drilling strategies
Micro-grid control system development and testing
Long-term performance monitoring (multi-year)